

```
[ec2-user@ip-172-31-16-54 ~]$ sudo mkdir linuxexam
[ec2-user@ip-172-31-16-54 ~]$ cd linuxexam
[ec2-user@ip-172-31-16-54 linuxexam]$ sudo touch student.txt
[ec2-user@ip-172-31-16-54 linuxexam]$ sudo touch course.txt
[ec2-user@ip-172-31-16-54 linuxexam]$ sudo touch result.txt
[ec2-user@ip-172-31-16-54 linuxexam]$ pwd
/home/ec2-user/linuxexam
[ec2-user@ip-172-31-16-54 linuxexam]$ cd
[ec2-user@ip-172-31-16-54 ~]$ tree linuxexam
linuxexam
├── course.txt
├── result.txt
└── student.txt

0 directories, 3 files
[ec2-user@ip-172-31-16-54 ~]$ |
```

```
[ec2-user@ip-172-31-16-54 ~]$ tree linuxexam
```

```
linuxexam
├── backup
│   └── student_backup.txt
├── course.txt
├── document
├── linux-course.txt
├── scripts
└── student.txt
```

```
3 directories, 4 files
```

```
[ec2-user@ip-172-31-16-54 ~]$ sudo nano linux_course.txt
[ec2-user@ip-172-31-16-54 ~]$ sudo cat student.txt
ROHIT PATIL
SIR VISVESRAYA INSTITUTE OF TECHNOLOGY NASHIK
AWS
DEVOPS
SQL
[ec2-user@ip-172-31-16-54 ~]$ sudo cat linux_course.txt
COURSE INCLUDE -
AWS
DEVOPS
SQL
PYTHON

[ec2-user@ip-172-31-16-54 ~]$ head -n 3 student.txt
ROHIT PATIL
SIR VISVESRAYA INSTITUTE OF TECHNOLOGY NASHIK
AWS
[ec2-user@ip-172-31-16-54 ~]$ tail -n 2 student.txt
DEVOPS
SQL
[ec2-user@ip-172-31-16-54 ~]$ head -n 3 linux_course.txt
COURSE INCLUDE -
AWS
DEVOPS
[ec2-user@ip-172-31-16-54 ~]$ tail -n 2 linux_course.txt
PYTHON

[ec2-user@ip-172-31-16-54 ~]$ wc student.txt
 5 11 74 student.txt
[ec2-user@ip-172-31-16-54 ~]$ wc linux_course.txt
 6  7 40 linux_course.txt
```

```
[ec2-user@ip-172-31-16-54 ~]$ sudo useradd student1
[ec2-user@ip-172-31-16-54 ~]$ sudo passwd student1
Changing password for user student1.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-16-54 ~]$ id student1
uid=1015(student1) gid=1017(student1) groups=1017(student1)
[ec2-user@ip-172-31-16-54 ~]$ grep student1 /etc/passwd
student1:x:1015:1017:./home/student1:/bin/bash
[ec2-user@ip-172-31-16-54 ~]$ su student1
Password:
[student1@ip-172-31-16-54 ec2-user]$ whoami
student1
```

```
[ec2-user@ip-172-31-16-54 ~]$ sudo groupadd linuxbatch
groupadd: group 'linuxbatch' already exists
[ec2-user@ip-172-31-16-54 ~]$ sudo groupadd linux_batch
[ec2-user@ip-172-31-16-54 ~]$ sudo usermod -aG linux_batch student1
[ec2-user@ip-172-31-16-54 ~]$ groups student1
student1 : student1 linux_batch
[ec2-user@ip-172-31-16-54 ~]$ sudo useradd student2
useradd: user 'student2' already exists
[ec2-user@ip-172-31-16-54 ~]$ sudo useradd student02
useradd: user 'student02' already exists
[ec2-user@ip-172-31-16-54 ~]$ sudo useradd student_2
[ec2-user@ip-172-31-16-54 ~]$ sudo usermod -aG linux_batch student_2
[ec2-user@ip-172-31-16-54 ~]$ getent group linux_batch
linux_batch:x:1018:student1,student_2
```

```
[ec2-user@ip-172-31-16-54 ~]$ sudo chmod 754 project.txt
[ec2-user@ip-172-31-16-54 ~]$ ls -l project.txt
-rwxr-xr--. 1 root root 0 Sep  4 17:21 project.txt
```

```
[ec2-user@ip-172-31-16-54 ~]$ sudo chmod 754 project.txt
[ec2-user@ip-172-31-16-54 ~]$ ls -l project.txt
-rwxr-xr--. 1 root root 0 Sep  4 17:21 project.txt
[ec2-user@ip-172-31-16-54 ~]$ sudo chmod 640 project.txt
[ec2-user@ip-172-31-16-54 ~]$ sudo chown student1 project.txt
[ec2-user@ip-172-31-16-54 ~]$ sudo chgrp linux_batch project.txt
[ec2-user@ip-172-31-16-54 ~]$ ls -l project.txt
-rw-r-----. 1 student1 linux_batch 0 Sep  4 17:21 project.txt
```

```
[ec2-user@ip-172-31-16-54 ~]$ sudo chmod 755 linuxexam/public
[ec2-user@ip-172-31-16-54 ~]$ sudo chmod 700 linuxexam/private
[ec2-user@ip-172-31-16-54 ~]$ sudo chmod 770 linuxexam/shared
[ec2-user@ip-172-31-16-54 ~]$ ls -l linuxexam/public
total 0
[ec2-user@ip-172-31-16-54 ~]$ ls -ld linuxexam/public
drwxr-xr-x. 2 root root 6 Sep  4 17:30 linuxexam/public
[ec2-user@ip-172-31-16-54 ~]$ ls -ld linuxexam/private
drwx-----. 2 root root 6 Sep  4 17:30 linuxexam/private
[ec2-user@ip-172-31-16-54 ~]$ ls -ld linuxexam/shared
-bash: ls-ld: command not found
[ec2-user@ip-172-31-16-54 ~]$ ls -ld linuxexam/shared
drwxrwx---. 2 root root 6 Sep  4 17:30 linuxexam/shared
```



```
[ec2-user@ip-172-31-16-54 linuxexam]$ grep "a" student_1.txt
anil 21 pune
shreya 22 mumbai
rahul 28 nashik
priya 21 pune
rohan 23 mumbai
anjali 20 pune
vikas 22 nagpur
karan 22 mumbai
pooja 23 pune
sakshi 20 nashik
[ec2-user@ip-172-31-16-54 linuxexam]$ grep -i "a" student_1.txt
anil 21 pune
shreya 22 mumbai
rahul 28 nashik
priya 21 pune
rohan 23 mumbai
anjali 20 pune
vikas 22 nagpur
karan 22 mumbai
pooja 23 pune
sakshi 20 nashik
[ec2-user@ip-172-31-16-54 linuxexam]$ grep -ic "a" student_1.txt
10
[ec2-user@ip-172-31-16-54 linuxexam]$ sort students.txt
sort: cannot read: students.txt: No such file or directory
[ec2-user@ip-172-31-16-54 linuxexam]$ sort student_1.txt
anil 21 pune
anjali 20 pune
karan 22 mumbai
pooja 23 pune
priya 21 pune
rahul 28 nashik
rohan 23 mumbai
sakshi 20 nashik
shreya 22 mumbai
vikas 22 nagpur
[ec2-user@ip-172-31-16-54 linuxexam]$ sort student_1.txt > sorted_student_1.txt
-bash: sorted_student_1.txt: Permission denied
[ec2-user@ip-172-31-16-54 linuxexam]$ find . -name "*.txt"
Find: './private': Permission denied
Find: './shared': Permission denied
Find: './shared': Permission denied
[ec2-user@ip-172-31-16-54 linuxexam]$ find . -name "*.txt"
./student.txt
./course.txt
./linux-course.txt
./backup/student_backup.txt
Find: './private': Permission denied
Find: './shared': Permission denied
./student_1.txt
[ec2-user@ip-172-31-16-54 linuxexam]$ cd
[ec2-user@ip-172-31-16-54 ~]$ find linuxexam -name "*.txt"
linuxexam/student.txt
linuxexam/course.txt
linuxexam/linux-course.txt
linuxexam/backup/student_backup.txt
Find: 'linuxexam/private': Permission denied
Find: 'linuxexam/shared': Permission denied
linuxexam/student_1.txt
```

```
[ec2-user@ip-172-31-16-54 ~]$ hostname
ip-172-31-16-54.ec2.internal
[ec2-user@ip-172-31-16-54 ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 0a:ff:e7:c6:55:fd brd ff:ff:ff:ff:ff:ff
    altname enp8s5
    altnam eni-029d86e47edb87bcf
    altnam device-number-0.0
    inet 172.31.16.54/20 metric 512 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 2620sec preferred_lft 2620sec
    inet6 fe80::8ff:e7ff:fec6:55fd/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
[ec2-user@ip-172-31-16-54 ~]$ ip linkk
Object "linkk" is Unknown, try "ip help".
[ec2-user@ip-172-31-16-54 ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 0a:ff:e7:c6:55:fd brd ff:ff:ff:ff:ff:ff
    altname enp8s5
    altnam eni-029d86e47edb87bcf
    altnam device-number-0.0
[ec2-user@ip-172-31-16-54 ~]$ ip route
default via 172.31.16.1 dev ens5 proto dhcp src 172.31.16.54 metric 512
172.31.0.0 via 172.31.16.1 dev ens5 proto dhcp src 172.31.16.54 metric 512
172.31.16.0/20 dev ens5 proto kernel scope link src 172.31.16.54 metric 512
172.31.16.1 dev ens5 proto dhcp scope link src 172.31.16.54 metric 512
[ec2-user@ip-172-31-16-54 ~]$ ping <ip>
-bash: syntax error near unexpected token 'newline'
[ec2-user@ip-172-31-16-54 ~]$ ping <IP>
-bash: syntax error near unexpected token 'newline'
[ec2-user@ip-172-31-16-54 ~]$ ping 34.229.152.20
PING 34.229.152.20 (34.229.152.20) 56(84) bytes of data.
^C
-- 34.229.152.20 ping statistics --
10 packets transmitted, 0 received, 100% packet loss, time 9358ms

[ec2-user@ip-172-31-16-54 ~]$ sudo ss -tulnp
Netid State Recv-Q Send-Q           Local Address:Port    Peer Address:Port    Process
udp    UNCONN 0      0           127.0.0.1:323         0.0.0.0:*             users:({"chronyd",pid=2283,fd=5})
udp    UNCONN 0      0           172.31.16.54%ens5:68  0.0.0.0:*             users:({"systemd-network",pid=1787,fd=17})
udp    UNCONN 0      0                :::1:323             [::]:*               users:({"chronyd",pid=2283,fd=6})
udp    UNCONN 0      0           0.0.0.0:546          [::]:*               users:({"systemd-network",pid=1787,fd=19})
tcp    LISTEN 0      128           0.0.0.0:22          0.0.0.0:*             users:({"sshd",pid=2172,fd=3})
tcp    LISTEN 0      128           [::]:22             [::]:*               users:({"sshd",pid=2172,fd=4})
tcp    LISTEN 0      511            *:80                *:80                  users:({"httpd",pid=1825,fd=4}, {"httpd",pid=1822,fd=4}, {"httpd",pid=1821,fd=4}, {"httpd",pid=1778,fd=4})
```

```
[ec2-user@ip-172-31-16-54 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:18:31 ago on Wed Aug 26 16:18:59 2026.
Package httpd-2.4.68-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-16-54 ~]$ sudo service httpd start
Redirecting to /bin/systemctl start httpd.service
[ec2-user@ip-172-31-16-54 ~]$ sudo service httpd status
Redirecting to /bin/systemctl status httpd.service
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─php-fpm.conf
   Active: active (running) since Wed 2026-08-26 16:18:54 UTC; 18min ago
     Docs: man:httpd.service(8)
  Main PID: 1762 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1059)
    Memory: 18.8M
       CPU: 1.136s
    CGroup: /system.slice/httpd.service
            └─1762 /usr/sbin/httpd -DFOREGROUND
              └─1803 /usr/sbin/httpd -DFOREGROUND
                └─1804 /usr/sbin/httpd -DFOREGROUND
                  └─1805 /usr/sbin/httpd -DFOREGROUND
                    └─1806 /usr/sbin/httpd -DFOREGROUND

Aug 26 16:18:54 ip-172-31-16-54.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Aug 26 16:18:54 ip-172-31-16-54.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Aug 26 16:18:54 ip-172-31-16-54.ec2.internal httpd[1762]: Server configured, listening on: port 80
[ec2-user@ip-172-31-16-54 ~]$ httpd -v
Server version: Apache/2.4.68 (Amazon Linux)
Server built:   Jun  8 2026 00:00:00
[ec2-user@ip-172-31-16-54 ~]$ sudo systemctl enable httpd
[ec2-user@ip-172-31-16-54 ~]$
```

student name - rohit

roll no - 53

course name - mca

'LINUX PRACTICAL EXAMINATION'